Warranty periods range from 2 to 20 years, but 5 years is typical for pavement warranties in the United States. Mitigation measures for deviations from acceptance criteria can be agreed to before beginning construction if they are not addressed in the specifications.

10.10.2 Monitoring Performance Parameters

Specific methods used to monitor performance are included in the specifications, along with requirements for when and how often measurements are to be taken. The owner generally makes these measurements, and it is critical that clear acceptance criteria and detailed descriptions of consequences of noncompliance are understood by all parties.

10.10.3 Dispute Resolution

Many warranty specifications have alternative dispute resolution terms for resolving disagreements. A method commonly used involves selection of a panel to evaluate and render an opinion on the specific item of the disagreement. The parties with the disagreement first must meet and clearly establish what the points of disagreement are. This keeps the panel on-track with the correct issues and reduces the amount of time needed to come to resolution. Each party selects one person who is independent and unaware of the dispute but has special knowledge or expertise relative to the issues disputed. A geotechnical engineer may be selected, for example, if the dispute is related to soil conditions. After each party selects one person, the two selected panel members meet and select a third. All three panel members agree to be unbiased and their costs are shared evenly. Information describing the disagreement and the position of each party is presented to the panel, which has an established time for evaluating the information. The panel then renders a decision, including any needed corrective action and an allocation of cost responsibility.

Resources

1. *A Summary of Operational Differences Between Nuclear and Non-Nuclear Density Measuring Instrument.* Sully-Miller Contracting Co., Quality Control Department. October 2000.
2. Burati, J.L., Weed, R.M., Hughes, C.S., and Hill, H.S. "Optimal Procedures for Quality Assurance Specifications." FHWA Report No. FHWA-RD-02-095, Office of Research, Development, and Technology, Federal Highway Administration, McLean VA, 1995.
3 D'Angelo, J. and Ferragut, T. "Summary of Simulation Studies from Demonstration Project No. 74 Field Management of Asphalt Mixes." Proceedings, Association of Asphalt Paving Technologists, vol. 60 (1991) pages 287–306.

4. *Hot Mix Asphalt Materials, Mixture Design and Construction,* 2nd ed., page 482. National Center for Asphalt Technology. NAPA Research and Education Foundation. Lanham, MD, 1996.
5. "Major Types of Transportation Construction Specifications: A Guideline to Understanding Their Evolution and Application." A report of the AASHTO Highway Subcommittee on Construction Quality Construction Task Force, Federal Highway Administration. August 2003.
6. NCHRP Synthesis 346. *State Construction Quality Assurance Programs: A Synthesis of Highway Practice.* National Cooperative Highway Research Program, Transportation Research Board. Washington DC, 2005.
7. *Quality Assurance Guide Specification.* American Association of State Highway and Transportation Officials. Washington DC, February 1996.
8. Sholar, Gregory A. et al. "Development and Refinement of the Florida Department of Transportation's Percent Within Limits Hot Mix Asphalt Specifications." Proceedings, Association of Asphalt Paving Technologists, vol. 75 (2006).
9. TRB Transportation Circular. *Glossary of Highway Quality Assurance Terms.*

CHAPTER 11

PAVEMENT MANAGEMENT

This chapter describes pavement management concepts, pavement evaluation, distress surveys and a pavement preservation program.

Section 11.1 Pavement Management Concepts
- 11.1.1 Introduction
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11.1 Pavement Management Concepts

11.1.1 Introduction

Transportation agencies have made significant investments in their pavement assets when both the costs of original construction and ongoing maintenance are considered. These assets are a vital part of the nation's economy, providing important links to markets, schools, medical facilities, and places of employment. Demand on this road network continues to grow with time, placing more importance on maintaining suitable pavement conditions at a time when increasing loads are causing pavements to deteriorate more rapidly.

Faced with these types of challenges, managers of transportation agencies are making use of tools that provide the type of information needed to support their decision-making processes. A Pavement Management System (PMS) is an example of this type of tool. The American Association of State Highway and Transportation Officials (AASHTO) defines PMS as: "A pavement management system is a set of tools or methods that assist decision-makers in finding optimum strategies for providing, evaluating, and maintaining pavements in a serviceable condition over a period of time." A PMS uses reliable pavement inventory and condition information to help identify and prioritize pavement maintenance and rehabilitation needs within budget or other constraints that may exist. A PMS may or may not be computerized, but a computerized PMS holds a significant advantage in the ability to manage and analyze large databases of pavement inventory information. Since most computerized PMSs are able to forecast future pavement conditions, an agency can quickly evaluate the impact of today's investment decisions on the long-term conditions of the road network. Consequently, agencies can use the information from a PMS to make more cost-effective maintenance and rehabilitation decisions, which results in a more efficient use of agency resources.

11.1.2 The Pavement Management Concept

Pavement management concepts can be used to manage the maintenance and rehabilitation of a road network by cost-effectively using a mixture of low-cost maintenance treatments on roads in good condition and more expensive rehabilitation treatments on roads with advanced levels of deterioration. This "mix of fixes" results in continuing efforts to reduce the number of roads in poor condition (which are very expensive to fix) while maintaining good roads in good condition for as long as possible.

Life-Cycle Cost Analysis

Determining the most cost-effective maintenance or rehabilitation strategy for an existing pavement can be a daunting task. However, a life-cycle cost analysis (LCCA) provides a means for comparing competing treatment strategies over an analysis period of 20 to 40 years. In an LCCA, all costs expected to be incurred over the life of a pavement are identified and converted to a single point in time using economic equations that represent the time value of money. Because all costs are converted to a single point in time, different treatment strategies with different performance lives can easily be compared. Additionally, an LCCA allows an agency to consider both agency costs and user costs in the same analysis. Agency costs are represented by direct costs to the agency for construction, maintenance, and rehabilitation. User costs, which are not always considered in an LCCA, represent the costs borne by the users under each scenario. For example, user costs might include differences in vehicle operating expenses, crash costs, or delay costs associated with the strategies being considered.

Good Roads Cost Less

A study conducted by the Utah Department of Transportation (UDOT) in 1977 demonstrated the concept that it costs less to maintain roads in good condition than it does roads in bad condition. Since that time, other agencies have used information such as that presented in Figure 11.1 to illustrate the differences in the relative magnitude of costs associated with maintaining roads in good condition versus those associated with allowing pavements to deteriorate until more substantial repairs are required. A cost-effective pavement management strategy optimizes the use of available maintenance

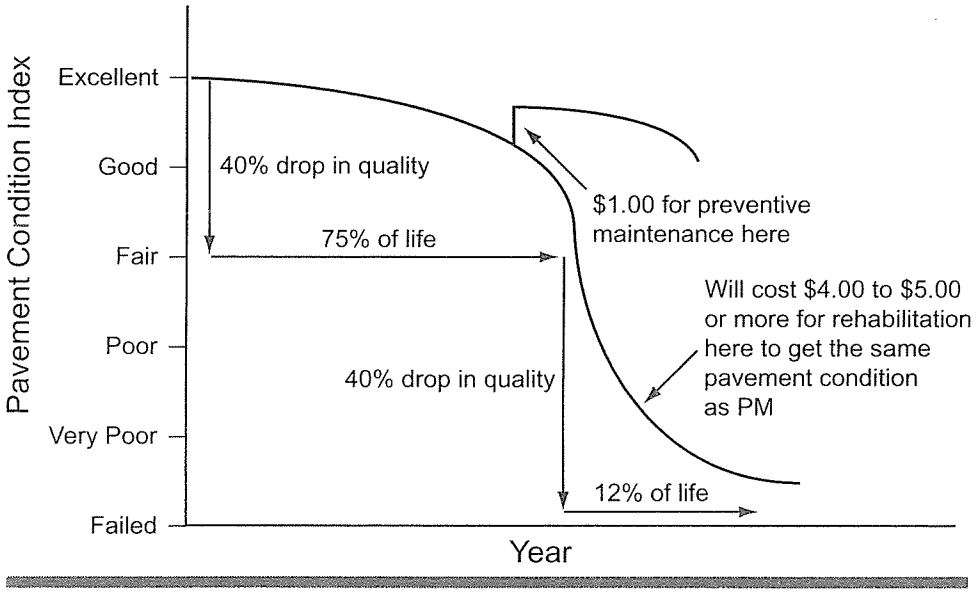


Figure 11.1 Typical Pavement Condition as a Function of Time

and rehabilitation funds, budgeting some funds to address roads requiring major rehabilitation or reconstruction while also budgeting funds for roads in good and fair condition to slow their rate of deterioration. By slowing the rate of pavement deterioration on these roads, an agency can maintain its road network at a higher overall condition level and defer the need for more costly rehabilitation treatments.

11.1.3 Pavement Management Components

Figure 11.2 illustrates the components of a typical pavement management system, and their interaction. A brief description of each component is described in the remainder of this section.

Inventory

To support a pavement management analysis, it is important that certain features of the road network be defined. Although the data elements that are needed may vary from agency to agency, most inventories contain the following types of information:

- Location information that includes the beginning and end point of each pavement facility
- Jurisdictional information, such as district, region, or county
- Road functional classification (e.g., arterial road)
- Road dimensions, including length, width, and number of lanes
- Historical information, such as original construction date, date of last treatment, and treatment history
- Surface characteristics, such as surface type

The development of the inventory can be a daunting task for agencies just starting in pavement management. Therefore, many agencies choose

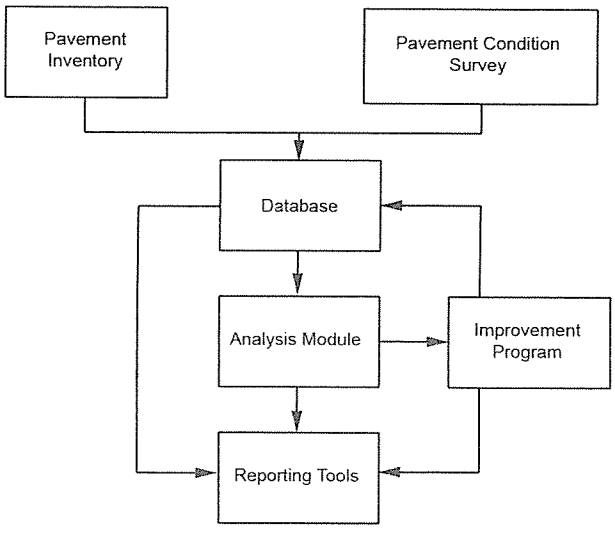


Figure 11.2 Typical Pavement Management Components

to use a two-step process to build their inventory. The first step is to add information that is important in making treatment decisions and is easy to obtain. For example, functional classifications or surface type information might be included in the first step. Over time, additional information that is less critical to the decision process, but helpful to have in a single location, can be added to the inventory. For example, construction history records might be added as the second step in building the inventory.

Condition Survey

A pavement management system uses the results of pavement condition surveys to help identify current and future maintenance and rehabilitation needs. All recommendations from the pavement management system are based on the results of the pavement condition survey so the importance of this component cannot be underestimated. The results of the survey are used to report current pavement conditions (usually in terms of a condition index), to forecast future conditions, and to determine appropriate maintenance and rehabilitation strategies.

The next section of this chapter discusses the different types of pavement condition surveys that are used to support a pavement management analysis. These techniques range from windshield surveys in which a subjective rating is assigned by a rater to represent the current pavement condition to more objective survey approaches in which distress type, severity, and quantity are recorded. Many agencies have begun using automated vehicles to collect pavement condition information. These vehicles operate at near-traffic speeds, collecting digital images of the pavement surface and using lasers to record pavement surface irregularities like roughness, faulting, and rutting.

Regardless of which approach is used, it is important that the survey procedures are followed consistently from one survey to another survey (and from one rater to another rater) and that they represent actual conditions. In general, objective condition surveys are more repeatable than subjective rating approaches, but they are often more labor intensive. Since it is important that pavement condition surveys are updated at least every three years, each agency must carefully evaluate the costs associated with each pavement condition survey procedure to ensure that the approach selected matches the resources available to support this activity.

Database

The inventory and condition information is stored in a pavement management database so it is available for use in generating reports and evaluating different maintenance and rehabilitation strategies. Databases can take a variety of forms, but relational databases are most commonly used in pavement management today.

Analysis Module

The heart of the pavement management system is the analysis module, where program recommendations for pavement maintenance and repair are

optimized within existing constraints, such as funding limitations. The most common methods of optimizing pavement maintenance and rehabilitation needs are described below:

- **Ranking.** A ranking formula is established giving higher weight to projects that reflect the agency's priorities. For instance, higher volume roads may have a higher priority than lower volume facilities.
- **Benefit/Cost Analysis.** Under this approach, the benefit associated with each feasible treatment (usually defined in terms of additional performance) is divided by the cost of the treatment to determine a benefit/cost ratio. Projects are prioritized so that those with the highest benefit/cost ratio are funded first.

To support the analysis, pavement performance models and treatment rules must be established. Pavement performance models describe the deterioration patterns of different types of pavements, using historical pavement condition data when available. In addition, treatment rules must be defined to describe the conditions under which each treatment is considered feasible and the expected cost of the treatment. For instance, a treatment rule might be established to apply an HMA overlay on roads in fair condition. These rules are used to identify all feasible treatment options for existing and predicted conditions during the analysis period. Finally, any budget constraints that limit the number of projects that can be funded each year must be defined.

Reporting Tools

The results of a pavement management analysis, and the information contained in the database, can be reported in a number of different formats, including text, tables, and graphics. Many agencies link their pavement management databases to geographic information systems (GIS) or other map packages to visually display the information. GIS displays of pavement management information are very effective tools when reporting pavement conditions to decision makers, elected officials, or the public.

Differences Based on Agency Size

Pavement management systems vary considerably in terms of the capabilities they provide, the user-friendliness of the program, and the degree to which the program can be customized to fit the particular needs of the agency. Both public domain and proprietary software packages are available. In general, public domain programs cost less to install but have limited capabilities and/or flexibility in meeting the needs of individual agencies. On the other hand, proprietary software packages are typically more expensive but also more sophisticated, with the flexibility needed to be tailored to meet the needs of an individual agency. When selecting pavement management software, an agency should first define its needs and then select a software package that meets those needs. Otherwise, the agency may incur expenses for features that are not needed and will not be used. For a pavement management system to be useful to an agency, it should addresses the agency's needs and be maintained over time according to the available resources.

11.1.4 Pavement Management Analysis Levels

Pavement management systems are used in transportation agencies to support decisions at three different levels: the project level, the network level, and the strategic level. Each of these levels is described further in this section of the chapter.

Project-Level Analysis

As the name implies, a project-level analysis involves decisions regarding the maintenance and rehabilitation of specific pavement sections. At this level, detailed pavement evaluations are conducted to determine the extent of any deterioration present, to identify the probable cause of the deterioration, and to design the appropriate treatment to correct the deficiencies. Pavement management systems can support decisions at this level by providing information on the deterioration rate of a pavement section, maintenance and construction histories, or other factors that may influence the selection of the appropriate repair strategy.

Network-Level Analysis

Pavement management systems are most commonly used at the network level, where the maintenance and rehabilitation needs of the entire road network are considered with respect to the amount of funding available. At the network level, the agency's maintenance and rehabilitation needs are prioritized so the most cost-effective use of available funding can be determined. Pavement needs are typically considered over a multiyear period and different maintenance and rehabilitation strategies are considered in the pavement management system. The initial recommendations from the pavement management system provide general guidelines for the type of work needed and possible locations where the work should be performed. The development of final work plans is usually an iterative process that involves moving neighboring projects from one year to another, or combining similar projects, to gain economy of scale. The work list is further developed at the project level in order to develop final project work plans and treatment designs.

Strategic-Level Analysis

In recent years, pavement management professionals have recognized the contributions of pavement management information to support an agency's strategic decisions. At this level, individuals such as governmental officials, upper agency management, and the engineers in charge of pavement maintenance, and bridge management all make decisions that play a part in determining pavement performance targets, establishing needed funding levels to achieve the performance targets, distributing funds among regions or districts, and establishing a pavement preservation policy. When strategic-level decisions must be made, there is a direct need to be able to utilize the information from the pavement management system in order to demonstrate the consequences on the long-term pavement condition of various treatment strategies, funding levels, or performance targets.

11.1.5 Role of Pavement Management in Pavement Preservation

One cost-effective strategy that an agency may consider for preserving its pavement network is a pavement preservation program that includes the use of low-cost preventive maintenance treatments to preserve the functional characteristics of a pavement in relatively good condition.

Definitions

The Federal Highway Administration's (FHWA) Pavement Preservation Expert Task Group defines a pavement preservation program as follows:

> A program employing a network level, long-term strategy that enhances pavement performance by using an integrated, cost-effective set of practices that extend pavement life, improve safety and meet motorist expectations.

These programs incorporate the use of preventive maintenance treatments, which have been defined by the AASHTO's Standing Committee on Highways as follows:

> A planned strategy of cost-effective treatments to an existing roadway system and its appurtenances that preserves the system, retards future deterioration, and maintains or improves the functional condition of the system (without significantly increasing the structural capacity).

Preventive maintenance treatments are typically applied to roads in relatively good condition before extensive amounts of deterioration are present. Examples of preventive maintenance treatments include crack sealing, chip seals, microsurfacing, and thin hot mix overlays.

Demonstrating Benefits of Pavement Preservation

Agencies that have initiated pavement preservation programs have used their pavement management systems to illustrate the expected benefits in terms of overall network conditions. For instance, the predicted overall network conditions associated with a pavement preservation program can be compared to another program in which a worst-first strategy is employed (where the worst roads in the network are fixed first), as illustrated in Figure 11.3.

The benefits of pavement preservation can also be demonstrated using the following techniques:

- A summary of treatment costs needed to achieve a targeted condition level using different strategies
- The expected distribution of network conditions in different condition categories (such as good, fair, and poor) under different preservation strategies
- A life-cycle cost comparison of the anticipated agency costs associated with a pavement preservation strategy as compared to a more traditional rehabilitation strategy

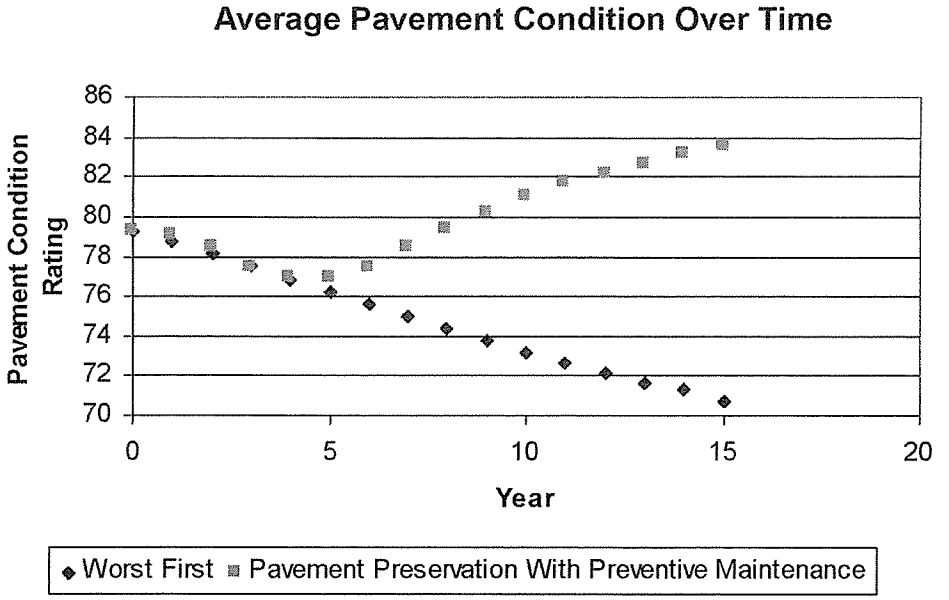


Figure 11.3 Average North Carolina Network Conditions Under a "Worst-First" as a Preservation Strategy

Selecting Treatments

A pavement management system can also be used to help identify good candidates for preventive maintenance. The results of pavement condition surveys can be used to identify roads that are not severely deteriorated or where structural deterioration has not taken place. Once a candidate for preventive maintenance has been identified, the section should be investigated in more detail to determine the extent of deterioration present and to verify that a preventive maintenance treatment is appropriate to correct the distress present or to prevent the further deterioration of the pavement surface. Further information about the appropriateness of preventive maintenance treatments to address pavement deterioration is provided elsewhere in this manual.

Long-Term Benefits to Pavement Preservation

Agencies that have strong pavement preservation programs in place have recognized a number of benefits, including those listed below:

- Higher customer satisfaction
- Better informed decisions
- Improved strategies and techniques
- Improved pavement condition
- Cost savings
- Increased safety

11.2 Pavement Evaluation

11.2.1 Introduction

A pavement management system relies on objective, repeatable pavement condition information to determine current and future maintenance and rehabilitation needs. Therefore, conducting a meaningful pavement evaluation is one of the first steps in the identification of feasible maintenance and rehabilitation strategies. Two types of pavement evaluations are normally conducted as part of an agency's pavement management practices: a functional evaluation and a structural evaluation. A functional evaluation considers the surface characteristics of a road, including certain types of cracking, surface smoothness, noise, and surface friction characteristics. A structural evaluation is used to determine the ability of the pavement structure to carry traffic loadings. A structural evaluation typically requires detailed information about pavement layer thicknesses, paving layer material properties, subgrade support conditions, traffic, and the response of the pavement to loading. Nondestructive testing (NDT) using falling weight deflectometer (FWD) devices is one of the most common approaches to determining the response of the pavement under traffic loadings. More detailed information on methods used to evaluate pavement functional and structural characteristics follows.

11.2.2 Functional Characteristics and Evaluation

Pavement surface characteristics are typically incorporated into pavement condition surveys to support the types of analyses conducted with a pavement management system. Three of the most common methods of evaluating the functional condition of a pavement are discussed here.

Pavement Distress

By collecting pavement distress information, an agency is able to determine the relative amount of pavement deterioration present and the appropriate level of repair that is needed to correct the deficiencies. Many different types of pavement distress are included in a pavement condition survey, as discussed further in Section 11.3. Some types of pavement distress (such as transverse and block cracking) reflect deterioration mechanisms that are more functional in nature, such as deterioration caused by climate or material properties. Other types of pavement distress (including alligator cracking or rutting) indicate load-related deterioration.

A comprehensive pavement distress survey provides a good indication of the overall condition of the pavement and suggests the primary mechanisms causing the deterioration.

Manual Surveys

Pavement condition surveys may be conducted using either manual or automated approaches. In a manual survey, individuals collect the pavement distress information by walking the pavement surface or by viewing the pavement surface from a slow-moving vehicle traveling along the shoulder. Manual surveys are more commonly used in areas without heavy traffic volumes, which might hinder the data collection process or impede the safety of the survey crews. An example of a manual pavement condition survey procedure for low-volume asphalt-surfaced roads is provided as Appendix E in the Asphalt Institute's MS-17, *Asphalt Overlays for Highway and Street Rehabilitation*. A copy is also available in the Asphalt Institute's Information Series, IS-169, *A Pavement Rating System for Low-Volume Asphalt Roads*.

Automated Surveys

Where traffic conditions must be considered, some agencies conduct automated surveys using data collection vehicles that capture digital or video images of the pavement surface at prevailing speed limits; these images are later processed in the office. In addition to addressing some of the safety considerations associated with the conduct of pavement condition surveys, many of the vehicles used to conduct automated surveys are also equipped with sensors to record rutting, roughness, and faulting measures. These vehicles may also include a camera for taking panoramic shots along the roadway, which can be useful for monitoring the condition of signs or other roadway hardware. Some devices are equipped with special lighting to eliminate problems caused by shadows or to facilitate surveys conducted at night. An example of an automated data collection van is provided in Figure 11.4.

The images captured during the survey process are typically post-processed in an office. Sensor data are usually processed in real time and reported in accordance with guidelines established by the agency. Distress data may be processed automatically using crack recognition software capable of recognizing and quantifying variations in grayscale that relate to the cracks on the pavement surface. Alternatively, the images may be viewed at a workstation where distress details are manually reviewed at a computer and distress quantities are recorded. This latter approach is referred to as a semi-automated method of distress processing.

The use of this technology has some disadvantages. The primary disadvantage is that it is difficult to discern characteristics associated with vertical displacement when using this technology. For example, it is difficult to determine whether there is a vertical displacement along a crack using this technology. Some pavement condition survey procedures use vertical displacement along a crack as a means of determining crack severity. It is also difficult to collect accurate sensor data in urban environments that prevent the equipment from operating at constant speeds. Because most agencies contract automated data collection services, an agency must also develop procedures to ensure the quality of the data collection and

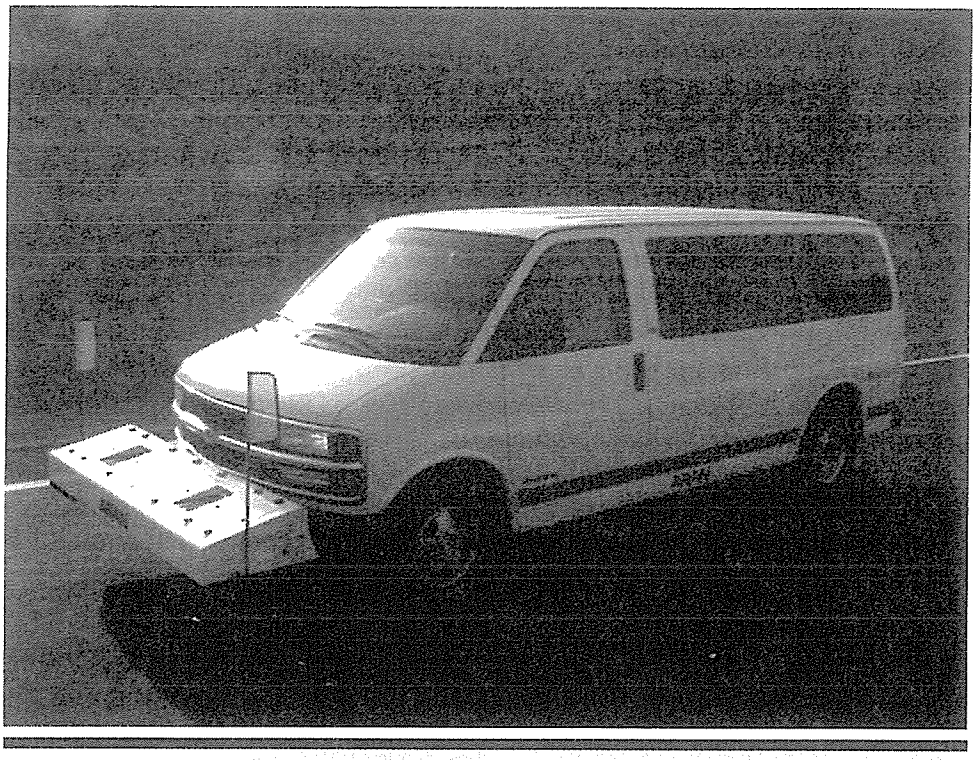


Figure 11.4 Automated Data-Collection Van

processing activities. The following types of considerations must be taken into account before using automated technology.

Sampling

Because of the resources required to conduct a pavement condition survey of an entire pavement network, some agencies utilize sampling techniques to determine the condition of each pavement section. Agencies that use this approach typically divide each pavement section into inspection units of a predetermined size, such as 2,500 square feet for asphalt-surfaced roads. A representative number of inspection units (or samples) are selected using statistical analyses. In general, a higher degree of sampling is required when a higher level of confidence is required or when there is considerable variability in the overall pavement condition.

Ride Quality

The longitudinal profile of a road provides an indication of the road's ride quality. Early measures of ride quality were based on the subjective ratings of passenger comfort within a vehicle traveling at posted speeds. Since that time, equipment has been developed to measure ride quality, which is typically reported in terms of an International Roughness Index (IRI). Ride quality, which is also referred to as roughness or smoothness, is a popular measure of a road's functional condition because it represents a condition measure that the public recognizes. The measurement of ride quality is common on highways and other high-speed facilities.

Types of Equipment

Most modern pavement-profiling devices measure vertical acceleration in response to the road and provide a pavement profile. Among the instruments on these vehicles are accelerometers that provide a horizontal plane of reference and sensors (laser, acoustic, or infrared) to measure the deviations of the pavement surface from the horizontal plane. These vehicles operate at near traffic speeds and are often outfitted with rut bars and cameras so that distress information can be collected at the same time. An example of this type of vehicle is provided in Figure 11.4.

Testing Approach

For acceptance testing on new construction, profilographs have been used in the past. Today, more agencies are using IRI for acceptance purposes. Profilographs produce a trace of the pavement surface, which is then compared with acceptable limits established by the agency. More information on the standard test method and equipment used to measure the pavement profile is available in ASTM E 1274, "Standard Test Method for Measuring Pavement Roughness Using a Profilograph."

Surface Friction

Adequate skid resistance, or surface friction, is an important functional characteristic that influences directional control and stopping ability on a wet or dry pavement. Most agencies become aware of surface friction problems during inclement weather, when the potential for hydroplaning (in which the vehicle tire has lost contact with the pavement surface and is actually riding on a thin film of water) is high. However, safety can also be compromised on dry pavement surfaces where surface friction is low.

Surface friction is a measure of the resistance provided by the pavement surface to the sliding of tires when they are prevented from rotating. In addition to pavement surface characteristics, surface friction can also be influenced by factors such as tire wear, vehicle speed, tire pressure, and the presence of water.

Surface friction information is typically collected as part of a safety program to reduce crash potential. However, agencies are beginning to incorporate surface friction measures into their pavement management systems to help identify areas that might benefit from treatments that restore pavement surface friction characteristics.

Types of Equipment

Three different approaches are used to measure surface friction: the locked-wheel mode, the slip mode, and the yaw mode. The most commonly used device to measure surface friction in North America is the locked-wheel skid trailer, such as the one shown in Figure 11.5. A test wheel is mounted on a trailer, which is pulled at the desired speed. Water is applied in front of the trailer, and the test wheel(s) is locked. The friction force at the tire and the speed of the vehicle are measured and reported in terms of a skid number (SN). The test is conducted in accordance with ASTM E 274, "Standard Test Method for Skid Resistance of Paved Surfaces Using a Full-Scale Tire," and either a standard ribbed or smooth (blank)



Figure 11.5 Locked-Wheel Skid Trailer

tire may be used, with the standard smooth tire providing a much better correlation between the surface texture characteristics and wet weather accidents.

Vehicles in slip mode operate on the premise that the highest friction factor, or critical slip, is found before the test wheel is locked. This phenomenon resulted in the development of the automatic control systems used on most aircraft today. Examples of slip devices include the Griptester and SAAB Friction Tester. Slip testing devices have seen widespread use on highway pavements in the United States.

The third approach to measuring surface friction is the yaw mode, which measures a sideways friction factor caused by a tire turned at an angle from the direction of motion. The Mu-Meter is an example of a yaw mode testing device. Another device for measuring sideways friction, the Sideways-Force Coefficient Routine Investigation Machine (SCRIM), has a test wheel mounted at 20 degrees to the direction of motion and operates at speeds of approximately 40 miles per hour.

Surface friction characteristics can also be estimated using macrotexture data provided as part of a pavement roughness evaluation. The results show good correlations to friction measures from more traditional approaches and increase the potential use of friction data as part of a network-level pavement management analysis.

Testing Approach

Friction data should be collected in the wheelpaths of each lane being tested. More detail about the equipment available to evaluate surface

friction characteristics and the test procedures that should be followed are available in ASTM E 274.

11.2.3 Structural Characteristics and Evaluation

A pavement structural evaluation is used to estimate the structural adequacy of an existing pavement to determine whether the pavement can sufficiently meet projected traffic loadings. In situations when the pavement is not adequate to meet projected loadings, the results of the structural evaluation can provide the information needed to design the appropriate treatment.

Because of the level of detail normally associated with this type of evaluation, structural evaluations are typically conducted as part of a project-level investigation. This may include more extensive pavement condition surveys conducted in conjunction with destructive testing (such as coring and boring) and with nondestructive testing using pavement deflection devices. Pavement deflection testing does not disturb the underlying pavement structure and does not require pavement materials to be removed for laboratory testing. However, it does require accurate pavement thickness information at each test location so coring is often conducted to verify pavement layer thicknesses.

Pavement Deflection Testing

Pavement deflection testing measures the response of the pavement to a load in terms of the resultant pavement deflection. The deflection data are used to backcalculate in situ material properties, to calculate pavement responses to load, and as inputs to other pavement analyses. By plotting maximum deflection measurements with road stations, areas of high deflections can be easily delineated to define project or subproject limits. The deflection information can also be used in deflection-based overlay design procedures.

Types of Equipment

Several types of pavement deflection devices are available, including impulse deflection devices (such as the falling weight deflectometer), steady-state dynamic deflection devices (such as the Dynaflect), and static deflection devices (such as the Benkelman Beam). The falling weight deflectometer (FWD) is the most commonly used deflection device for assessing pavement responses.

The FWD is capable of simulating in magnitude and duration the dynamic loads applied to a pavement by wheel loads. A dynamic load is applied by dropping a set of weights onto a circular load plate. A range of load magnitudes can be achieved by varying the number of weights used and the heights from which they are dropped. The actual load magnitude is measured using a load cell within the load plate, and the resulting deflections are measured by transducers located at the center of the load

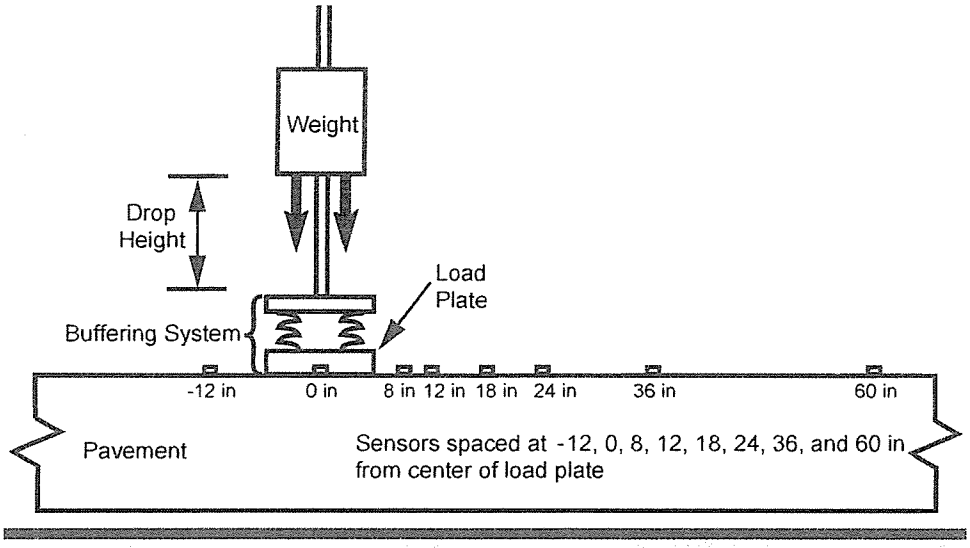


Figure 11.6 Schematic Diagram of FWD Testing Equipment

plate and at preset distances from the load plate. This is illustrated in the schematic in Figure 11.6. Both the maximum surface deflection under the load and the magnitude of the deflection basin are used to provide information on the structural capability of the pavement structure.

Rather than apply an impulse load, steady-state dynamic deflection devices apply a large static preload to the pavement, causing a sinusoidal vibration that is measured in terms of amplitude and frequency. The magnitude of the load that can be applied using this type of device is limited by the amplitude of the dynamic force that is created. Additionally, the preload may alter the pavement properties so that the resulting deflections may not be representative of the deflections that would occur under a moving wheel load. For these reasons, this type of equipment is not used as extensively as the FWD.

The original deflection devices applied a static load to the pavement surface and measured the resulting deflections. The most well-known static deflection device is the Benkelman Beam, which was developed in the early 1950s at the WASHO Road Test. The beam is placed on the pavement surface with its probe located between the dual tires of a loaded truck. The rebound of the pavement as the truck moves away is measured and reported. This device has also been replaced by the FWD.

Testing Approach

Pavement deflection testing requires traffic control because the testing device must be stopped at each test location. On flexible pavements, deflection testing is typically conducted in the outer wheelpath, but a testing configuration may test both the inner and outer wheelpath at staggered intervals. Additional information on recommended FWD testing frequencies is provided in ASTM D 4694, "Standard Test Method for Deflections with a Falling-Weight-Type Impulse Load Device."

11.2.4 Use of Pavement Condition Information

The results of a functional pavement evaluation provide the information needed for the pavement management system to identify and prioritize maintenance and rehabilitation needs. Typically, pavement distress and roughness information are used to calculate pavement condition indexes that can be used to compare the condition of one pavement section with another and to identify the level of repair that might be needed to address the deficiencies identified. Over time, pavement condition information helps establish rates of pavement deterioration that can be used to forecast future pavement conditions under different funding scenarios.

Pavement Condition Indexes

To facilitate the use of pavement distress information in a pavement management system, pavement condition indexes are calculated based on the type, severity, and quantity of distress present. Pavement condition indexes typically range in value from 0 to 10 or 0 to 100, with the highest values corresponding to pavements in excellent condition. Points are subtracted from a perfect score based on the distresses observed in the field to calculate the index. The complexity of the pavement condition index calculation varies depending on the type of survey being conducted.

Pavement condition indexes can be calculated for each type of distress present or a composite index can be used that aggregates all of the distress into one index representing the overall condition of a pavement section. Some agencies use a combination of these two types of indexes; the individual indexes are used to identify appropriate maintenance and rehabilitation treatments and the composite index is used to report the overall network conditions.

In local agencies, a composite index is most commonly used. For example, the pavement rating system for low-volume asphalt pavements that is included in several Asphalt Institute publications requires the rater to assign a value between 0 and 5 for all functional distress present (such as longitudinal and transverse cracking) and 0 to 10 for distress representing structural forms of deterioration (such as alligator cracking and rutting) or other important deterioration mechanisms (such as deficient drainage or poor ride quality). The defects associated with each distress are summed and subtracted from 100 to determine the condition rating for a pavement section.

A more detailed approach to calculating a composite index was developed by the U.S. Army Corps of Engineers for use in its Micro PAVER pavement management system. This approach, which is documented in ASTM D 6433, "Standard Practice for Roads and Parking Lots Pavement Condition Index Surveys," uses a similar approach, but separate deduction values have been developed for each distress type, severity, and quantity calculation.

Composite indexes are very useful for communicating the overall condition of a section and for comparing conditions of multiple sections. In addition, a composite index provides an indication of the overall level of repair needed, as shown in Figure 11.7.

In addition to using a composite index, some agencies use individual indexes or raw distress information to identify feasible maintenance and rehabilitation treatments. For example, fatigue cracking, patching, and rut distress may be combined to determine a structural condition index while longitudinal and transverse cracking may be combined with block cracking to form a functional condition index. Pavement sections with a low structural condition index are not good candidates for preventive maintenance treatments although a section with low functional condition indexes and high structural indexes might be.

Reporting Current Conditions

The most common use of pavement condition information is to report overall network conditions. This information is reported in a number of different formats depending on the needs of the intended recipients. For some individuals, a report that presents the overall network conditions by functional class, surface type, or some other delineator may be appropriate. Several agencies regularly publish reports on the state of the road network after each data collection cycle. Other agencies publish limited pavement condition information on their web sites.

One of the most effective ways of communicating pavement condition information involves displaying pavement condition information on color-coded maps. This approach has a strong visual impact, especially when roads in poor condition are shown in red. This method of displaying pavement condition information is especially useful for presentations to elected officials and upper management.

One of the newer methods of reporting pavement conditions is in terms of remaining service life (RSL). An RSL is an estimate of the number of years until the pavement reaches a threshold level at which the pavement condition is so poor that the only feasible treatment option is

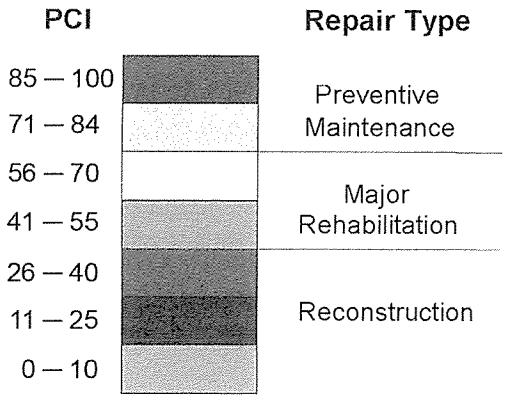


Figure 11.7 Sample Link between Pavement Condition Index (PCI) and Repair Type

reconstruction. An RSL of 0 indicates that a pavement section has reached the threshold condition and no longer functions as it was originally intended. Overall network conditions may be reported in terms of an average RSL or by the percent of the network in different RSL categories (such as 0–2 years, 3–5 years, 5–10 years, 10–15 years, and greater than 15 years). As with a pavement condition index, feasible maintenance and rehabilitation treatments can be determined for each RSL category.

When reporting pavement conditions, it is important that the information be conveyed in a manner that can be quickly understood. For this reason, many agencies choose to report network conditions in terms of condition categories such as good, fair, and poor. To report pavement conditions in this manner, an agency must define each condition category in terms of its pavement condition indexes. For example, in Figure 11.7 a PCI between 70 and 100 might represent a road in good condition, a PCI between 40 and 70 might represent a road in fair condition, and a PCI less than 40 might represent a road in poor condition. Any number of condition categories can be used, depending on the needs of the agency.

Forecasting Future Conditions

Once a pavement condition index has been developed, statistical tools can be used to develop pavement performance models that reflect the rate at which the pavements deteriorate over time. The inclusion of models that forecast future pavement condition is one of the most important factors differentiating a pavement management system from a pavement database. Prediction models allow an agency to forecast future pavement conditions under any number of budget scenarios or under different treatment strategies so that the long-term impacts of these options can be more fully understood and compared.

11.3 Pavement Distress Surveys

Whether part of a formal pavement management system or a project-level pavement evaluation to select an appropriate maintenance or rehabilitation treatment, pavement distress surveys provide information on the type and the extent of deterioration present and the primary mechanisms causing the distress. These considerations are both very important in selecting and designing an improvement. Pavement distress surveys may be conducted in conjunction with other testing, such as destructive or nondestructive testing.

11.3.1 Distress Survey Approaches

The previous section introduced methods of conducting both functional and structural pavement evaluations. Pavement distress surveys are the most common type of functional pavement evaluation used by highway and public works agencies. As part of a network-level evaluation, a distress survey may be conducted on selected samples to provide an overall condition for each pavement section in the network. As part of a project-level investigation in which a rehabilitation treatment is being designed, more detailed distress surveys are typically conducted. At the project level, distress maps showing distress location, severity, and extent are often created. Distress maps are time-consuming to prepare but provide an excellent way of estimating repair quantities or marking the location of drainage or geometric features. An example of a distress map is shown in Figure 11.8.

Depending on the types of distress survey being conducted, the following types of equipment may be required:

- A distress identification manual
- A map or utility layout sheet showing section limits
- A measuring wheel
- Survey forms and/or mapping sheets
- A clipboard
- Pencils
- A straightedge or string line
- Spray paint (to mark limits and section identifiers)
- A hand calculator
- A camera or other recording device

Pavement distress surveys are typically conducted using crews of at least two people so that while one individual measures distress quantities, the other serves as a recorder. Having two crew members is also important for safety reasons because survey crews often have to mingle with traffic.

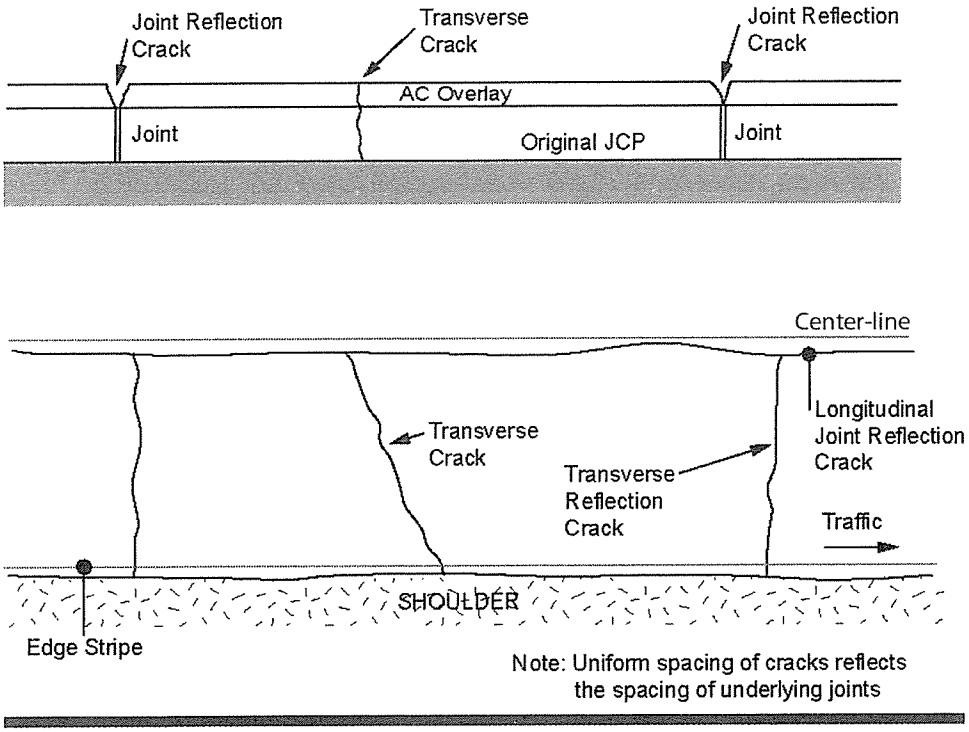


Figure 11.8 Sample Distress Map

To help ensure the reliability of a pavement distress survey and to minimize variations in survey results between raters, survey crews should follow the established survey procedures as closely as possible. Detailed distress surveys require raters to correctly differentiate between distress types and to use prescribed definitions outlined in a distress identification manual to distinguish between different severity levels. Distress identification manuals also provide guidance on how to measure different types of distress and often provide tips for recording a distress that is difficult to discern in the field.

Before beginning a pavement distress survey, each rater should be trained in accordance with the distress identification procedures to minimize discrepancies in the field. In addition to being able to identify each distress type included in the survey procedures, the raters should be familiar enough with the procedures to know whether all severities associated with a particular distress should be collected or whether only the highest severity is recorded. Other details, such as how distress in a patched area should be handled, ought to be discussed before the start of the survey. These considerations are important for maintaining consistency in the survey data since raters may differ in what they consider a "good" road.

Examples

A number of pavement distress survey approaches can be used for both network- and project-level surveys. One of the most well-documented survey procedures is the Pavement Condition Index (PCI) survey procedure developed by the U.S. Army Corps of Engineers and described briefly earlier in this chapter. The PCI, which is documented in ASTM D 6433 for

roads and parking lots, includes 19 distress types for hot mix asphalt roads and 19 for Portland cement concrete roads. For the majority of distress types, three severity levels are defined.

The Federal Highway Administration (FHWA) developed its "Distress Identification Manual for the Long-Term Pavement Performance Program" to support the data collection efforts associated with Long-Term Pavement Performance (LTPP) studies initiated in 1987. This manual has been through several iterations, with the current version issued in 2003 (FHWA 2003). Designed as a project-level inspection procedure, the manual identifies distress types and severity levels for three pavement surfaces: asphalt concrete surfaces, Portland cement concrete surfaces, and continuously reinforced concrete surfaces. The manual serves as an excellent resource for project-level and research surveys; however, an approach for calculating a distress index from the distress data has never been developed. Without an index calculation, the procedure has limited application for network-level pavement condition surveys.

The Asphalt Institute's Manual, MS-17, includes condition evaluation forms for flexible and rigid pavements used by the Ontario Ministry of Transportation (after some editing for consistency with Asphalt Institute terminology). These forms illustrate the level of detail often associated with a pavement distress survey.

Characteristics of an Effective Survey

An effective pavement distress survey should have the following characteristics:

- There should be little interpretation expected of the raters in the field.
- The methodology should be objective and repeatable.
- It should minimize incursions with traffic.
- It should provide a sufficient level of detail so the agency can estimate maintenance and rehabilitation needs accurately.
- It should include a method of estimating distress quantities.
- It should differentiate distress severity levels in situations where distress severity changes dramatically over a relatively short period of time.
- The importance of the distress in determining the appropriate repair option should be factored into the calculation of the pavement condition index.

11.3.2 Typical Asphalt Pavement Distress Types

Before the appropriate type of repair can be determined, the type and extent of deterioration in a pavement section must be understood and the cause of the distress identified. This section categorizes common types of distress into the five general categories listed here:

- Cracking
- Distortion
- Disintegration